



NAME:
HABIB ID: SECTION:
INSTRUCTOR:

MATH 101 CALCULUS 1

FALL 2023

TEST 1

INSTRUCTIONS

Duration: 50 minutes

- This test has **5 questions** worth a total of **25 marks**. There are **5 printed pages** and **1 blank page** for rough work (please count these now!)
- Attempt **all** questions.
- Show **all** relevant working in the space below each question. Generally, correct answers with no working

1. (a) Using implicit differentiation, find the derivative of $y = \tan^{-1}(x)$: [2]

$$\begin{aligned} y = \tan^{-1}x &\Rightarrow x = \tan y \\ \frac{dx}{dx} &= \frac{d}{dx}(\tan y) \Rightarrow 1 = \sec^2 y \frac{dy}{dx} \\ \frac{dy}{dx} &= \frac{1}{\sec^2 y} \end{aligned} \quad \left| \quad \begin{aligned} \therefore \sec^2 \theta &= 1 + \tan^2 \theta \\ \Rightarrow \frac{dy}{dx} &= \frac{1}{1 + \tan^2 y} \\ \Rightarrow \boxed{\frac{dy}{dx} = \frac{1}{1 + x^2}} \end{aligned} \right.$$

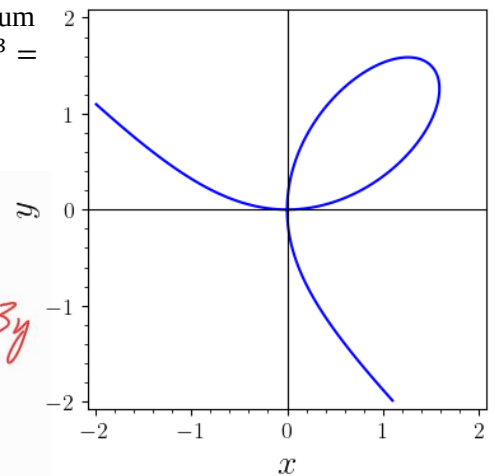
- (b) Using the formula that you derived in part (a), find the derivative of $y = \arctan(\sqrt[3]{x^2})$: [3]

$$\begin{aligned} \text{Let } u &= x^{2/3} \Rightarrow y = \arctan(u) \\ \frac{du}{dx} &= \frac{2}{3} x^{-1/3}, \quad \frac{dy}{du} = \frac{1}{1+u^2} = \frac{1}{1+x^{4/3}} \\ \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \Rightarrow \boxed{\frac{dy}{dx} = \frac{2}{3x^{1/3}(1+x^{4/3})}} \end{aligned}$$

2. The figure on the right shows the graph of the curve known as the Folium of Descartes (from Latin *folium* 'leaf'), defined implicitly by $x^3 + y^3 = 3xy$.

(a) Find the critical point on the curve where $x > 0$.

[3]



$$\begin{aligned}
 x^3 + y^3 &= 3xy \quad (1) \\
 \text{a) Differentiating both sides: } 3x^2 + 3y^2 y' &= 3xy' + 3y \\
 \Rightarrow y'(3x^2 - 3y^2) &= 3x^2 - 3y \\
 \Rightarrow y' &= \frac{3x^2 - 3y}{3x^2 - 3y^2} = \frac{x^2 - y}{x^2 - y^2} \\
 y' = 0 \text{ when } x^2 - y &= 0 \Rightarrow y = x^2 \\
 \text{Taking } y = x^2 \text{ in (1) gives } x^3 + x^6 &= 3x^3 \\
 \Rightarrow x &= \sqrt[3]{2}
 \end{aligned}$$

- (b) Using your work from part (a), apply the second derivative test to show that the point found above is a local maximum.

[3]

$$\text{b) } y'' = \frac{(x - y^2)(2x - y') - (x^2 - y)(1 - 2yy')}{(x^2 - y^2)^2}$$

$\rightarrow = 0$
 $\rightarrow > 0$

Sign of y'' must be < 0 for it to be a max. point. So, with $x > 1$,

$$\begin{aligned}
 \text{Sign}[y''] &= \text{Sign}[(x - y^2)] = \text{Sign}[x - x^4] \\
 &< 0
 \end{aligned}$$

3. Given $y = ax^n$, where n is a positive number prove that $\frac{dy}{dx} = anx^{n-1}$ using the definition of derivative:
[4]

Using definition of the derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \rightarrow \textcircled{1}$$

$$f(x) = ax^n \rightarrow f(x+h) = a(x+h)^n \rightarrow \text{expand using binomial expansion}$$

$$(\therefore (a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_n a^0 b^n)$$

$$f(x+h) = x^n + nx^{n-1}h + {}^nC_2 x^{n-2}h^2 + \dots + h^n$$

subst. $f(x)$ and $f(x+h)$ in $\textcircled{1}$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{a(x^n + nx^{n-1}h + {}^nC_2 x^{n-2}h^2 + \dots + h^n) - ax^n}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a(n x^{n-1} h + {}^nC_2 x^{n-2} h^2 + \dots + h^{n-1})}{h} = \lim_{h \rightarrow 0} a(n x^{n-1} + {}^nC_2 x^{n-2} h + \dots + h^{n-2})$$

Applying limit

$$= anx^{n-1} + 0 + 0 + \dots + 0$$

$$f'(x) = anx^{n-1} \quad \text{where } n > 0$$

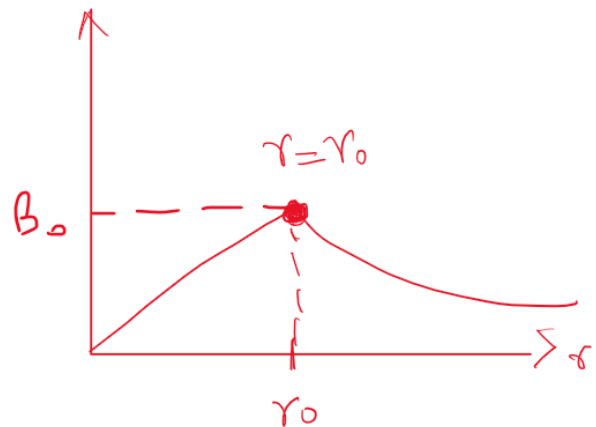
4. A magnetic field, B , is given as a function of the distance, r , from the center of a wire as follows:

$$B = \begin{cases} \frac{r}{r_0} B_0, & \text{for } r \leq r_0 \\ \frac{r_0}{r} B_0, & \text{for } r > r_0. \end{cases}$$

(a) Sketch a graph of B against r .

[1]

$$B = \begin{cases} \frac{B_0}{r_0} (r) & , r \leq r_0 \\ B_0 r_0 \left(\frac{1}{r} \right) & , r > r_0 \end{cases}$$



(b) By finding the relevant two-sided limits, decide whether B is continuous at $r = r_0$. Explain.

[3]

$$\lim_{r \rightarrow r_0^-} B(r) = \lim_{r \rightarrow r_0^-} \frac{B_0}{r_0} (r) = \frac{B_0}{r_0} \lim_{r \rightarrow r_0^-} (r) = \frac{B_0}{r_0} r_0 \Rightarrow \boxed{\lim_{r \rightarrow r_0^-} B(r) = B_0}$$

$$\lim_{r \rightarrow r_0^+} B(r) = \lim_{r \rightarrow r_0^+} B_0 r_0 \left(\frac{1}{r} \right) = B_0 r_0 \lim_{r \rightarrow r_0^+} \left(\frac{1}{r} \right) = \frac{B_0 r_0}{r_0} \Rightarrow \boxed{\lim_{r \rightarrow r_0^+} B(r) = B_0}$$

$$B(r_0) = \frac{B_0}{r_0} (r_0) \Rightarrow \boxed{B(r_0) = B_0}$$

$$\text{Since, } \lim_{r \rightarrow r_0^-} B(r) = \lim_{r \rightarrow r_0^+} B(r) = B(r_0),$$

the function is continuous at $r = r_0$.

(c) Is B differentiable at $r = r_0$? Explain.

[2]

Not differentiable at $r=r_0$ because the graph of B has a sharp corner/turn at $r=r_0$.

For checking differentiability: $LHD = RHD$

$$\underline{LHD}: \frac{d}{dr} \left(\frac{r}{r_0} B_0 \right) \Rightarrow \frac{B_0}{r_0} \left(\frac{dr}{dr} \right) = \frac{B_0}{r_0} (1)$$

$$\underline{RHD}: \frac{d}{dr} \left(\frac{r_0}{r} B_0 \right) \Rightarrow r_0 B_0 \frac{d}{dr} \left(\frac{1}{r} \right) = r_0 B_0 \left(\frac{-1}{r^2} \right) \Big|_{r=r_0} = \frac{-r_0 B_0}{r_0^2} = -\frac{B_0}{r_0}$$

Since $LHD \neq RHD$, function is not differentiable at $r=r_0$

5. Differentiate the following functions.

(a) $f(x) = \ln(e^x x^3)$

[2]

$$\text{let } u = e^x x^3 \Rightarrow f(u) = \ln u$$

$$\frac{du}{dx} = e^x x^3 + 3x^2 e^x$$

$$\frac{du}{dx} = e^x x^2 (x+3)$$

$$f'(x) = \frac{df}{du} \cdot \frac{du}{dx}$$

$$f'(x) = e^x x^2 (x+3) \times \frac{1}{u}$$

$$f'(x) = \frac{e^x x^2 (x+3)}{e^x x^{3-2}}$$

$$\boxed{f'(x) = \frac{x+3}{x}}$$

(b) $y = 4^{4x^4}$

[2]

$$\text{let } u = 4x^4 \Rightarrow \frac{du}{dx} = 16x^3$$

$$\Rightarrow y = 4^u \Rightarrow \frac{dy}{du} = \ln 4 \cdot 4^u$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$\frac{dy}{dx} = \ln 4 \cdot 4^u \times 16x^3$$

$$\frac{dy}{dx} = \ln 4 \cdot 4^{4x^4} \cdot 16x^3$$

$$\frac{dy}{dx} = \ln 4 \cdot 4^{4x^4+2} x^3$$